

How Information Travels Across a Network

TCP/IP Model — Secure Online Banking Access (HTTPS)

ICDFA E-Campus — Group Project

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Presentation: 12–15 minutes

1. Scenario: Opening a Secure Website

Scenario: a customer opens the Access Bank online banking portal over HTTPS.

The client first uses DNS to resolve the online banking service name to an IP address.

The browser establishes a TCP connection to the banking web service using destination port 443.

TLS then establishes a secure session before protected application data is exchanged.

Traffic crosses the customer's local network, router/firewall, Internet routers, and the banking service infrastructure.

The response returns through the network and is decapsulated as it reaches the customer's device.

Note: This is an illustrative packet journey based on a real-world online banking scenario; it does not claim to expose the bank's internal network.

2. The Four Layers of the TCP/IP Model

Application

HTTPS, HTTP, DNS — network services used by applications

Transport

TCP/UDP — ports, reliability and end-to-end delivery

Internet

IP — logical addressing and routing

Network Access

Ethernet/MAC — local-link delivery and framing

3. TCP/IP to OSI Model Mapping

TCP/IP	OSI	Corresponding functions
TCP/IP Application	OSI 5–7	Session, Presentation, Application
TCP/IP Transport	OSI 4	Transport
TCP/IP Internet	OSI 3	Network
TCP/IP Network Access	OSI 1–2	Physical + Data Link

4. Application Layer: DNS and HTTPS

DNS translates the online banking service name into an IP address.

The browser creates an HTTPS request after learning the destination address for the banking service.

HTTPS uses HTTP with TLS protection, providing encryption and server authentication for the banking session.

Application data is passed to the Transport layer.

At the receiver, the processed data is delivered back to the web application/browser.

5. Transport Layer: TCP and Port 443

TCP is connection-oriented and provides reliable, ordered delivery.

Port numbers identify the communicating service; HTTPS commonly uses TCP port 443.

The TCP three-way handshake is SYN → SYN-ACK → ACK.

Sequence numbers and acknowledgements support reliable delivery.

After TCP is established, TLS negotiation can protect the application session.

6. Internet Layer: IP Addressing and Routing

IP provides logical addressing so packets can travel between the customer's network and the banking service.

The source IP identifies the client and the destination IP identifies the banking service endpoint.

Routers examine destination IP addresses and select the next hop.

Off-subnet traffic normally goes to the client's default gateway.

A firewall/router may perform NAT on outbound traffic.

An IP packet can cross multiple routers before reaching the banking service network.

7. Network Access Layer: MAC and Ethernet

Ethernet frames carry IP packets across a local network.

MAC addresses identify network interfaces on the local link.

ARP can discover the MAC address associated with a local IPv4 address.

At a router hop, the old Ethernet frame is removed and a new frame is created.

Therefore MAC addresses can change hop-by-hop while the destination IP remains the end-to-end target.

8. Encapsulation and Decapsulation

Application data — HTTPS/TLS

TCP segment — TCP header + data

IP packet — IP header + segment

Ethernet frame — Ethernet
header/trailer + packet

Sender: encapsulation ↓

Network: forwarding across links

Receiver: decapsulation ↑

Data reaches the browser

9. Original Packet Journey: Customer to Online Banking Service



Outbound: application data → TCP segment → IP packet → Ethernet frame → routed to the server.

Return traffic travels back toward the client and is decapsulated layer-by-layer.

10. Protocol Analysis Table

Layer	Protocol	PDU	Addressing	Purpose
Application	DNS	Message	Domain / server IP	Name resolution
Application	HTTPS/TLS	TLS records / HTTP data	Server IP + TCP 443	Secure web communication
Transport	TCP	Segment	Source/destination ports	Reliable delivery
Internet	IP	Packet	Source/destination IP	Routing
Network Access	Ethernet	Frame	Source/destination MAC	Local-link delivery

11. TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	ACKs, sequencing, retransmission	No built-in delivery guarantee
Ordering	Ordered byte stream	No ordering guarantee
Overhead	Higher	Lower
Examples	HTTPS, SSH	DNS, VoIP/media

12. Communication Failure and Troubleshooting

Failure example: the banking service name resolves, but the HTTPS connection cannot be established.

1. Check the client's IP address and default gateway.
2. Test connectivity to the local gateway.
3. Test DNS resolution to separate name-resolution problems from connectivity problems.
4. Test TCP port 443 and inspect firewall logs.
5. Capture traffic in Wireshark and look for SYN, SYN-ACK and ACK.
6. If SYN packets leave but no response returns, check routing, firewall policy, NAT and upstream connectivity.
7. Correct the fault, retest HTTPS and confirm successful TLS/application traffic.

References

RFC 9293 — Transmission Control Protocol (TCP), IETF/RFC Editor.

RFC 9110 — HTTP Semantics, IETF/RFC Editor.

RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3, IETF/RFC Editor.

RFC 1034 / RFC 1035 — Domain Name System, IETF/RFC Editor.

OPNsense Documentation — Networking, firewall and NAT documentation.

Cisco Networking Academy — Networking fundamentals and TCP/IP concepts.